Forgetting Is a Certificate: A Verified Developmental Lifecycle for Continual Learning

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Abstract

Continual-learning methods promise not to forget; none of them can prove it. We present a system in which retention is a *checked invariant* rather than a score: a network grown as an **organism** of frozen parts ("organs"), each behind a finite, checkable **contract** (class anchors, margins, and a proximity territory), routed by first-accept. Growth happens exactly where the organism abstains; the parent is never touched; and in every run of every experiment—twelve benchmarks across three substrates—growth caused **exactly zero answer changes**, verified bitwise. The guarantee's price is then engineered down through a biologically ordered lifecycle: *adolescence* (gate-verified consolidation trained purely on contract-sampled data), learned *inhibition* (vetoes trained on the contracts' own samples), and a late-maturing prefrontal *arbiter* restricted to organs whose contracts accept the input, so that every answer of the final system is contract-covered—measured at an uncertified fraction of 0.0000 across all 36 evaluation runs. Lifecycle changes are admitted by an acceptance gate that we audit and then confirm on seeds never used for any decision: it admits no regression above its own stated resolution, refuses every harmful change it can resolve, and refuses no clearly-helpful one. On held-out evaluation seeds against deliberately strengthened baselines, the certified organism beats same-data replay on five of twelve tasks with zero stored samples (split-MNIST 0.893 vs. replay's 0.871) and reaches 0.835 strict on CIFAR-100 at SimpleCIL's own ViT-B/16 backbone via contracted prototypes ($+2.2$ over SimpleCIL-last, -8.7 to RanPAC); the certified lifecycle closes the strict-guarantee gap on CIFAR-100 from 18 points to 4. An unconstrained arbiter would close it to 1, but 94% of its increment comes from answering outside its own certified territories, so we report it as an ablation, not a result. No configuration beats a properly trained joint model, and we retract our development-time "beats joint" observations. Negative results are reported with their mechanisms and, where we found them, their laws: routing headroom bounds what any router can win; duplicate organs buy coverage, never precision; monotone boundary edits cannot replicate selection. We argue that auditable, refusable change—not accuracy—is the axis on which continual learning should also be scored.

1 Introduction

Two recent lines of work each solved half of a problem. The first [our companion developmental study; 1, 2] showed that in growing networks it is *training, not pruning*, that destroys what a network was born with, and that trajectories which touch fewer weights retain more—but its retention was only ever a measurement. The second, on *interval contracts* at network interfaces, showed that a boundary inside a network can carry a finite, checkable specification sufficient to re-derive the module behind it without data—but closed the book on growth: on both its substrates, verified structural growth accepted zero candidates, because training destroys the error structure growth needs.

Each hole is the other's theorem. If parts are frozen behind contracts, the protective mechanism of the first line (weights never touched are never overwritten) becomes an *invariant* of the second; and if growth adds a *fresh representation* behind a contract rather than anchors inside an old one, the closure of the second line dissolves. This paper builds that synthesis and pushes it through a full developmental lifecycle.

Contributions. (1) *The organism* (§3): frozen contracted organs in a first-accept cascade, grown where the organism abstains. Retention is a theorem, and it is *checked, not trusted*: bitwise, in every run, growth changes no answer. (2) *The contract as its own negative sampler* (§3): every unsupervised proximity geometry we tested saturates at the same false-accept frontier; the lever that breaks it is supervision the organism already owns—vetoes trained on contract-sampled positives in whitened territory coordinates against real falsely-accepted samples. (3) *The certified lifecycle* (§4): adolescence (verified consolidation, trained data-free from the contracts), and—mirroring the posterior-to-anterior maturation of cortex [3]—a prefrontal *arbiter* over frozen parts. We measure exactly where an unconstrained arbiter's accuracy comes from, find that much of it is bought *outside* every certified territory, and adopt the territory-respecting variant, so the final system's every answer is contract-covered. (4) *The acceptance gate* (§5): every lifecycle change is admitted by a gate combining a data-free territory check with a paired statistical test on a real held-out escrow; we audit it against every historical decision and confirm it on seeds never used for any decision. (5) *The price list and its laws* (§6, §7): the guarantee ladder priced across twelve benchmarks and two backbones, and three findings—routing headroom, the coverage-only value of redundancy, the impossibility of replicating selection by monotone boundary edits—shown to be one mechanism. (6) *The failure modes* (§10), each with a mechanism, none tuned away.

2 Related work

Continual learning. Regularization methods [4] collapse in the single-head class-incremental setting [5]; replay methods [6, 7] work by storing data or a generator; generative classifiers [8] are the strongest exemplar-free baseline at MNIST scale. In the pretrained regime, prototype methods dominate [9, 10, 11]. Architecturally our organism descends from Expert Gate [12] and progressive networks [13]; none of these provide verified retention, audited change events, abstention semantics, or replaceable parts, which is the axis this paper adds. **Certified training.** Our contracts reuse the machinery of certified robustness [14] pointed inward at interfaces rather than outward at perturbations; the companion contracts study found the two guarantees do not imply each other. **Reservoirs and interference.** The developmental companion connects reservoir computing [15, 16] to catastrophic interference [17]: sparse trajectories are implicit weight protection. Here the protection is made structural.

3 The organism

An **organ** is a small prototypical trunk with spectrally capped layers behind a **ball contract**: per-class feature anchors with margin slack τ and radius ρ, plus a learned input-space *territory*—per-class PCA subspaces (means, axes, eigenvalues, calibration scalars; prototypes, never raw samples) with cap ρ_{in}. An organ *accepts* an input only inside its territory and with a determined margin; accepted predictions carry a certified robust radius by arithmetic. The **organism** is a first-accept cascade of organs frozen at birth. Its abstention region is the *need field*: growth trains a fresh organ only on what the organism declines, and a candidate joins only if it verifiably improves the declined validation mass (certificate parameters are solved, never penalized). Throughout this paper, *contract-covered* means an answer produced by an organ whose contract accepted the input—a routing property checked exactly, distinct from the per-answer robust-radius certificate that accepted answers carry in addition.

Retention invariant. Under growth, the routing prefix and every organ's decision depend only on parameters frozen at birth, so any answered input keeps its answer under all future growth; coverage is monotone. With monotone *tightening* (accepted sets only shrink), the invariant is *no silent drift*: an answer changes only by falling through to a later organ at an explicit, logged event. We do not trust this: every experiment compares answers bitwise before and after every growth event. The count, across twelve benchmarks, three substrates, all seeds and all lifecycle configurations, is zero.

Learned inhibition. The organism's failure mode is the false accept: an organ confidently claiming classes it never saw. A shoot-out of five unsupervised proximity geometries (centroids, per-class PCA, a capped autoencoder, combinations, a corruption discriminator) shows they plateau at the same false-accept frontier. The lever is supervision the organism already owns: a contract is a data generator, so when task B arrives, each old organ gains a **veto**—a small capped net trained on contract-sampled positives *in the territory's whitened coordinates* (where $z \sim \mathcal{N}(0, I)$ is exactly the model the spec asserts) against the real falsely-accepted samples, conjoined with the accept rule. Exclusion is 0.955–0.986 at 6–10% self-loss; the raw-space version learns "synthetic vs. real" and rejects everything (self-loss 1.000)—a vacuous certificate. An ablation shows scalar threshold-tightening becomes *net harmful* once vetoes exist (its self-loss releases samples that downstream organs answer wrongly): none 0.408, scalar 0.742, veto-only 0.897, both 0.819 on split-MNIST.

Spatial substrate. On a spatially embedded recurrent sheet (positions in $[0, 1]^2$, hinge wiring cost, cost-weighted pruning), organs are local recurrent populations placed by the wiring economy; growth is *directional* (a newborn may listen to frozen predecessors; no edge ever enters a frozen row). Lesioning organ k destroys its own tasks (damage 0.96–0.98) and no earlier task (lower-triangular damage 0.000 across all organisms); listening carries $+8.7$ points of forward transfer on temporal streams and its cost is visible as lesion fragility. Anatomy is modular by measurement (within-organ weight 0.93 vs. a 0.25 null) and *replaceable*: an organ re-derived from its published contract alone, at three architectures, agrees with the original at 0.92–0.93 (swapped-spec control: 0.000).

4 The certified lifecycle

Development's invariant is also its ceiling: frozen errors accumulate. Biology answers with staged consolidation—sensory cortex first, prefrontal control last, plastic over already-frozen parts before consolidating itself [3]. The organism affords the same stages because every part carries a generator.

Adolescence. Sample labeled data from every organ's contract, train one consolidated organ on the pooled synthetic set, and *verify*: a usefulness floor on holdout coverage (vacuous consolidations pass every safety check—measured), and a *relative* per-class gate against the organism being replaced on identical data (absolute floors harden with class count). At high class counts, consolidation must be into *closed-form prototypes*: the trained-trunk variant collapses on a 100-class pool (0.40 raw, worse with more capacity). The consolidated organ inherits the *merged* territory (a union of PCA territories is a PCA territory), which keeps it veto-capable—this single property turns mid-life adolescence from harmful (-5 points; the unvetoable version wrongly claims 19% of later classes) to net-positive ($+6$; residual channel 4–5%). Contract *guarantees* are renegotiated freely at adolescence and certified backward-compatible by sampling (coverage kept 1.000, agreement 0.93–1.00); *territories* cannot be re-derived from generated data—every attempt is quietly vacuous (the renegotiated organ answers 0–7% of real inputs and the organism silently reverts).

The prefrontal arbiter, and where its accuracy comes from. A small net reads every organ's evidence (distance, margin, territory score, accept bit) and learns *who to trust*. It answers nothing; it routes. It trains purely on contract-sampled data and stays plastic while the parts are frozen. Left unconstrained, it answers with its argmax organ *whether or not that organ accepted the input*, and an exact additive decomposition of its gain over first-accept (partitions: *fill* where the cascade abstained, *re-route* where it answered, *drop* by θ-abstention) shows what that freedom buys. On digits the gain is genuine re-routing (134 corrections against 4 regressions over five seeds), but 8.6% of its answers come from a non-accepting organ and carry 45% of the advantage; on CIFAR-ViT, where 100 fine-grained territories are a poor guide to who should answer, outside-territory answers are 15.6% of the total and carry **94%** of the advantage—only 18.3% of its re-routes land on an accepting organ, its abstention threshold never binds, and it has stopped behaving like a contracted system at all. The unconstrained arbiter is therefore not a gateable mechanism but an *uncertifiable* one (§5 confirms this by audit), and the lifecycle adopts the **territory-respecting arbiter**: the same net, its choice restricted to organs that accept, abstaining when none do. Every answer of the final system is then contract-covered by construction—measured at exactly 0.0000 uncertified answers across all evaluation runs—at a price §6 states plainly: the restriction keeps 72% of the unconstrained gain on digits and 32% on CIFAR-ViT, where the regime-appropriate remedy is topological (contracted

prototypes, certified by construction, match the unconstrained arbiter's accuracy). The arbiter is consolidated last behind the same relative gate, its answer changes audited as one logged event.

5 The acceptance gate

Every lifecycle change—consolidation, arbitration, renegotiation, death—is admitted or refused by a gate. Our original gate scored per-class accuracy on contract samples plus the last task's real validation, and two changes walked past it: the unconstrained arbiter (accepted $3/3$ while buying 94% of its CIFAR gain outside every territory) and a boundary renegotiation that was quietly -0.0045 on real test (§7). Both slipped through for one reason: **gates that score contract samples certify contract fidelity, not real behaviour**. The repaired gate adds two checks and keeps the original, so each verdict names what fired.

Territory—data-free and exact. Every answer must come from an organ that *accepted* the input: no data, no labels, no threshold. It also repairs a representational gap: the verdict reports `certified` separately from `accepted`, making "beneficial but uncertified"—the honest description of the unconstrained arbiter—sayable at all. Audited against all 21 historical arbiter decisions, this check refuses every one of them, harmful and helpful alike (1.3–9.8% of any unconstrained arbiter's answers are outside territory), so it is a *policy*, not a detector: it prevents the wall-robot false-accept (§10) as a matter of category, at the price of every beneficial unconstrained arbiter. Detecting a regression *as such* needs real data.

Escrow—a paired test on real held-out data, used only by gates, never for training. Two designs failed first and both are traps worth naming. A fixed 0.05 per-class floor over 100 classes at $\sim$60 samples each *false-refused* a territory-clean change worth $+0.024$, because the minimum of 100 noisy per-class estimates falls below any small fixed slack almost surely. Drawing the escrow from *train* flattered every candidate (a genuinely -0.0048 change scored $+0.0003$), because organs fit train better than test. And capping the escrow at N per class silently discarded most of a small dataset's validation split, collapsing a decisively refusable regression ($z = -6.86$ on the full split) to an accepted one ($z = -1.57$). The final check uses the *whole* held-out validation split, judges overall quality by a one-sided McNemar test on discordant pairs ($z \geq -1.645$; only discordant pairs carry information), flags a class only past a Bonferroni correction, and keeps an unconditional hard floor at 0.25 so a single-class collapse—the apoptosis failure mode—stays catchable regardless of significance. In the audit it refuses all four harmful historical arbiters and none of the eleven helpful ones (one false refusal appears only at full-split power, on a $+0.014$ change flagged per-class); its resolution is reported by the gate itself (`detectable_drop`), so it degrades by stating its limit rather than failing silently.

Escrow independence. The escrow shares the validation split with the lifecycle's own gates. Splitting validation into class-stratified halves—one for the gates, one for the escrow—changes 2 of 30 evaluation-seed verdicts, both toward *more* acceptance (one -0.015 regression and one $+0.005$ gain now admitted); none flips from accept to refuse, which is what reuse-inflated acceptance would produce. The reuse is benign and the cost of independence is power (half the escrow is $\sim\sqrt{2}$ less), so the frozen gate keeps the whole split.

Disclosure and out-of-sample confirmation. The organism and lifecycle were frozen before evaluation seeds 2–4 ran, but the gate was not: both escrow redesigns responded to evaluation-seed failures, so the gate's in-sample record (zero accepted regressions over 36 runs) required confirmation. We froze the specification above and reran the ten CPU benchmarks on **seeds 5–7, never used for any decision**: over 30 fresh runs the uncertified fraction stays 0.0000, wall-robot's harmful arbiter is refused $3/3$ ($z = -3.35, -6.34, -7.72$), no clearly-helpful candidate is refused, and exactly one accepted change regresses—satimage seed 6 at -0.0068, *half* the gate's stated resolution for that escrow (0.0137 at n=723), a validation-to-test gap no test on this data could resolve. The out-of-sample guarantee therefore takes its precise form: **the frozen gate admits no regression above its stated resolution, refuses every harmful change it can resolve, and refuses no clearly-helpful one.** Across all 102 runs of the frozen gate now on record (evaluation, confirmation, and two class-order permutations) the exceptions are two sub-resolution slips and one genuine miss—letter under a permuted order, -0.023 at $z = -1.10$, where validation did not show a regression that test does—so

Table 1: Class-incremental learning, single head, no task oracle, abstention counted as error for the organism ("strict"). Evaluation seeds 2–4; baselines strengthened (z-scored features, doubled training, doubled-width joint). ^{L} = after the *certified* lifecycle (territory-respecting arbiter, repaired gate): every answer contract-covered, zero accepted regressions. 2–3 seeds. †Shared-speaker split; under a speaker-disjoint split every method loses roughly half its accuracy (organism 0.35, replay 0.45, joint 0.52; §10).

benchmark	fine-tune	EWC	replay	**organism**	joint
split-digits (5×2)	0.58	0.58	0.93	0.88 (**0.91**^{L})	0.98
split-MNIST (5×2)	0.25	0.39	0.87	**0.89**	0.98
split-KMNIST (5×2)	0.27	0.34	0.77	**0.81**	0.96
split-Fashion (5×2)	0.21	0.32	0.78	**0.80**	0.89
letter (13×2)	0.04	0.10	0.50	**0.78**	0.95
satimage (3×2)	0.33	0.33	**0.88**	0.83	0.90
wall-robot (2×2)	0.23	0.36	0.72	**0.80**	0.91
sheet, synthetic streams	—	—	0.99	0.98–0.99^{L}	1.00
sheet, spoken digits†	—	—	0.93	0.77 (0.79^{L})	0.95
CIFAR-100, frozen R18	—	—	0.59	0.40 (0.47^{L})	0.67
CIFAR-100, ViT (lifecycle)	—	—	0.86	0.67→**0.79**^{L}	0.81
CIFAR-100, ViT (contr. protos)	—	—	—	**0.84**	0.84 (NCM)

the statement holds with a measured exception rate of one in 102. CIFAR confirmation seeds require growing fresh organisms and are deferred to the GPU stage of this project.

A lifetime budget. Sub-resolution slips can compound over a long life of gated events, so the gate also carries a *ledger*: every accepted change records its signed escrow delta, and a candidate that passes its own test is refused if the cumulative delta would fall below -0.02. On the sleeping organism (§8; a gated consolidation and a gated dream every night, seven MLP benchmarks, evaluation seeds) the escrow check alone refuses 25 nights the original contract-sample gate would have judged blind, and the ledger binds exactly where predicted and nowhere else: on the one long life (letter, 13 nights) accepted drift accumulates to -0.013 and a single dream is refused for the budget; on every five-night life cumulative drift stays ≈ 0 or positive. It changes no final accuracy here—its value is on long deployments, where accumulation is the only way a correctly-gated system can still drift.

6 Results

The guarantee ladder, priced. On CIFAR-100 the same features support three regimes: bitwise answer-stability (cascade), frozen-prototypes with logged drift, and no guarantee. The strict guarantee costs $\approx$16 points on weak features, about the same on strong ones (0.667 vs. 0.844 on evaluation seeds), and ≈ 0 on the MNIST ladder—the exchange rate is set by how well the features separate classes. The logged-drift rung costs nothing versus free NCM and adds a working certificate: drift is 5–$17\times$ more often corrective than harmful and is *forecastable before each event* from contract samples alone on the promised region (correlation 0.65–0.98, calibration 1.5–$2\times$).

The certified lifecycle, and what certification costs. On CIFAR-ViT evaluation seeds: growth $0.667 \rightarrow$ adolescence $0.776 \rightarrow$ territory-respecting arbiter 0.794 ($3/3$ seeds, all gates passing), with every answer contract-covered; contracted prototypes remain the strongest fully-covered configuration at 0.835 strict / 0.844 forced, so our CIFAR-ViT recommendation is the prototype topology rather than cascade-plus-arbiter. That recommendation is substrate-dependent. On three stronger frozen caches (ViT-B/16 on the full train set, DINOv2-B, ViT-L/16; evaluation seeds, frozen gate) the contracted prototypes' forced accuracy equals nearest-class-mean on the same features to the third decimal (0.842 / 0.873 / 0.877) and the strict number trails it by the 0.8–1.0 point of abstention the contract buys: the certificate rides feature quality one-for-one but never above the prototype topology it wraps, so the 8.7-point gap to RanPAC on ViT-B is its trained random-projection head, not the features. DINOv2-B gives the best certified CIFAR-100 number, 0.865, and there the cascade route equals the prototype route (0.865 vs. 0.865) because the free arbiter's headroom over the cascade

Table 2: Against published results, where standard protocols exist. All "this work" numbers are the certified configuration; the unconstrained arbiter's higher numbers appear in parentheses as ablations. MNIST-scale anchors are the class-IL benchmark runs of van de Ven and Tolias [5], van de Ven et al. [8] (iCaRL and DGR are their implementations of Rebuffi et al. [7] and Shin et al. [6]); both ViT-scale anchors are *final* (last-step) accuracies, the same metric as our strict score.

setting	best published	**this work,**
split-MNIST, exemplar-free	gen. classifier 0.938 [8]; DGR 0.913 [6]	0.893 (0.9
split-MNIST, with buffer	iCaRL 0.946 [7] (2,000 exemplars)	0.893, zero
split-MNIST, same-data replay	strengthened replay 0.871	**0.893**, zero
CIFAR-100, ViT-B/16-in21k	SimpleCIL 0.813 last [9]; RanPAC 0.922 [11]	**0.835** / 0.8
CIFAR-100, DINOv2-B	no verified class-IL anchor; same-feature NCM 0.873, same-data replay 0.889	**0.865** / 0.8
CIFAR-100, ViT-L/16-in21k	no verified anchor; NCM 0.877, replay 0.892	**0.867** / 0.8
ImageNet-R, ViT-B/16-in21k	SimpleCIL 0.546; APER-adapter 0.680 [9]; RanPAC 0.779 [11]	**0.524** / 0.5
ImageNet-A, ViT-B/16-in21k	SimpleCIL 0.489; APER-adapter 0.494 [9]; RanPAC 0.624 [11]	**0.454** / 0.5
CUB-200, ViT-B/16-in21k	SimpleCIL 0.867; APER-adapter 0.871 [9]; RanPAC 0.903 [11]	**0.858** / 0.8
ObjectNet-200, ViT-B/16-in21k	SimpleCIL 0.536; APER-adapter 0.580 [9]; RanPAC not reported	**0.506** / 0.5
OmniBenchmark, ViT-B/16-in21k	SimpleCIL 0.732; APER-adapter 0.744 [9]; RanPAC 0.799 [11]	**0.724** / 0.7
VTAB, ViT-B/16-in21k	SimpleCIL 0.844; APER-adapter 0.844 [9]; RanPAC 0.922 [11]	**0.888** / 0.8
CIFAR-100, from scratch	DER 0.653 last [18]; FOSTER similar [19] (trained backbone, 2,000-exemplar buffers)	not compa

has shrunk from $+0.059$ to $+0.008$; the gate refuses the arbiter in 8 of 9 runs, each time by the last-task validation clause, forgoing $+0.001$ to $+0.007$, and accepts one change of $+0.001$, so its record stays at zero accepted regressions over 45 evaluation runs. Same-data replay still leads by 2.5–3 points on every backbone. On the six-dataset pretrained-model suite those papers report on with the same backbone (ImageNet-R, ImageNet-A, CUB, ObjectNet, OmniBenchmark, VTAB; features extracted on a GPU, pipeline unchanged, seeds 2–4) the finding repeats without exception: forced equals nearest-class-mean to the third decimal on all 18 seeds and lands within about one point of SimpleCIL's last-step accuracy on five datasets (Zhou et al.'s supplementary Inc20/Inc30 tables, IN21K block), while RanPAC's head buys a further 4–23 points. The certified lifecycle trails the prototypes by 0.2–10.6 points in proportion to how much of the input the consolidated organ answers (91–100% on VTAB, ImageNet-A, CUB; 43–75% on ImageNet-R, ObjectNet, OmniBenchmark), and the gate refused all 18 arbiter proposals, 11 of them on the escrow's real-data test. The suite also exposed three latent defects in code that had only ever seen full-width territories (a shape error for classes with fewer than 25 samples; NaN anchors and NaN-poisoned gate aggregates for classes with almost no data), fixed without touching any threshold and with no effect on any earlier number. We also correct one anchor: the CIFAR-100 APER-adapter value we quoted earlier (0.852) was the B0 Inc5 column; at B0 Inc10 it is 0.875. Finally, two curricula in the developmental sense (a different protocol, compared only to the random-order lifecycle of the same seeds) locate where scaffolding matters. Contracted prototypes are order-invariant by construction; only the cascade moves. Ordering CIFAR-100 tasks by superclass (confusable classes together) raises growth from 0.667 to 0.698 and the certified lifecycle from 0.794 to 0.799, while easy-first, hard-first and spread orders stay within the order band. Growing 20 superclass *containers* first and refining each with a child organ inside its territory reaches 0.626 fine strict (0.696 forced) with 30 vetoes instead of 250, zero fine-level flips across ten refinements, and a certified coarse answer on 20% of inputs; but it cannot reach the flat prototypes, because a 20-way nearest-mean over superclasses scores 0.849 on these features against 0.911 for the fine means mapped to superclasses, a loss no refinement recovers (hierarchical NCM ceiling 0.777 vs. 0.843). Scaffolding pays in inhibition, stability and graded answers, not in accuracy; the unit that helps accuracy is task composition, not a coarse label. Prototype children (closed-form anchors, no trunk) leave fine accuracy essentially unchanged (0.665 vs. 0.626) at a fraction of the parameters; two-step refinement inside containers needs 138 within-container vetoes and produces no within-container flips; and consolidating progressively every two tasks ends at 0.771 against 0.776 for one consolidation at the end, with every proposal gated. The lever itself was then measured: a RanPAC-style random-projection ridge head placed under

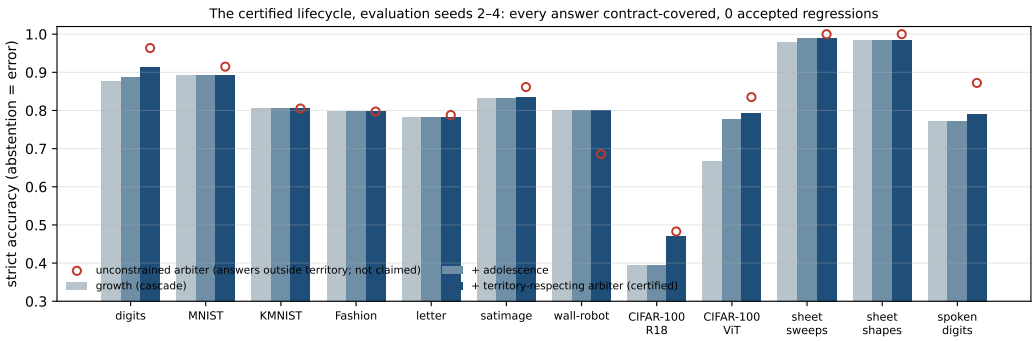


Figure 1: The certified lifecycle on all twelve benchmarks (evaluation seeds 2–4, strict accuracy): growth, adolescence, and the territory-respecting arbiter under the repaired gate. Every bar is contract-covered; hollow markers show the unconstrained arbiter, whose extra accuracy is bought outside certified territory and is not claimed. Refused stages carry the previous stage forward, which is why several bars are flat.

the same contract (per-class PCA territory, margin on the head's top-two scores) moves certified strict accuracy on CIFAR-ViT from 0.835 to 0.875 and on DINOv2-B from 0.865 to 0.901, within a point of RanPAC's head-only row on CIFAR and level with its full result on VTAB (0.929), while coverage stays above 0.96 everywhere but ImageNet-A, where a second-order head has too few samples per class and the contract refuses accordingly. What the head gives up is the cascade's invariant: already-given test answers change 8–155 times per 100 test points over a run because the ridge weights are re-solved, against exactly zero for the cascade; the contract restores coverage semantics but not retention. We then tested whether the two can be had at once. Freezing the head the way an organ is frozen – one ridge block per task, solved from that task's rows alone and never re-solved, answers taken first-accept in birth order – delivers the invariant exactly: appending a block changes zero answers, checked bitwise on every seed. It does not deliver the accuracy. Against the re-solved head it loses 38.7 points on CIFAR-100 (0.482 vs. 0.869), 20.5 on DINOv2-B and 54.1 on VTAB, falling below even the contracted prototypes. The reason is structural rather than an implementation gap: a ridge readout's advantage over class means is that it is solved over all classes jointly, and a block frozen at birth has no negatives for classes that do not yet exist. Retention requires the answer to be fixed when first given; joint decorrelation requires knowing every class. We therefore report accuracy and bitwise retention as separate configurations, with the reason named. Three further measurements price the configurations (§5 protocol, seeds 2–4). Cutting the same CIFAR-100 stream into 20, 50 and 100 updates leaves the prototypes' answers identical to twelve decimals and the cascade's growth flips at zero over 501 checked steps, while the re-solved head ends at the same accuracy and rewrites $13 \to 41$ answers per 100 test points along the way ($40 \to 141$ on ImageNet-R); the certified lifecycle falls from 0.795 to 0.729 at 100 organs, and the escrow refused a candidate 14 points worse than its incumbent on every seed there. As an out-of-distribution score the certificate slack ranks CIFAR-10 against CIFAR-100 at AUROC 0.841, textures (DTD) at 0.995 and out-of-scope text against CLINC150 intents at 0.955 – above kNN, NCM-cosine and Mahalanobis on the same features but below a jointly trained probe's softmax – yet at its own operating point it still accepts 79% of CIFAR-10 and 99% of SVHN digits, which a frozen encoder maps inside object territories: the certificate bounds where an answer came from, not how far outside an input lies. After the shared encoder the prototypes cost 1.9 M floats, the cascade 4.9 M, and the M$=10{,}000$ head 10.6 M plus a 101 M-float Gram to keep learning; the two configurations that never change an answer carry no learning state at all. The headline table replicates on its own untouched seeds 5–7 (all seven datasets, first-session adapters): the contracted head moves by at most 1.4 points, growth flips are 0 on 21/21 seeds, the gate accepted 3/21 arbiters with one sub-resolution regression (-0.4, CUB) and no helpful refusal, and the CIFAR arbiter that was accepted 3/3 on the evaluation seeds is refused 3/3 here at the rule's threshold. On RanPAC's own checkpoint family the head with adapters

reaches forced 0.777 / 0.607 / 0.895 on ImageNet-R / -A / CUB against RanPAC's 0.779 / 0.624 / 0.903, the remaining difference being the metric definition. The certificate transfers unchanged to a frozen sentence encoder (Banking77, CLINC150: head 0.921 / 0.950 strict, above same-data replay and a joint MLP, flips 0) and to raw intrusion-detection rows (NSL-KDD: head 0.935, prototypes 0.428 because class means do not separate that space, every consolidation refused, correctly). Finally we ran the one ingredient the published leaders have and we did not: a single parameter-efficient tuning of the backbone on the first task, frozen thereafter (rank-16 adapters, 0.31M parameters against the backbone's 86M). Read in our terms this is a birth step, so it costs no guarantee: with it the contracted head reaches 0.906 on CIFAR-100 (APER-adapter 0.875, RanPAC 0.922), 0.763 on ImageNet-R (APER 0.680), 0.639 on ObjectNet (best published APER variant 0.582) and 0.939 on VTAB, above RanPAC's full result there, with coverage 0.97–0.99 and growth flips still exactly zero. The exception is instructive: on OmniBenchmark the birth step *lowers* feature quality (-0.012 joint nearest-class-mean) and the acceptance gate then refused the consolidation that follows it on all three seeds, so the anatomy stayed as grown. Extracting the three datasets whose published RanPAC numbers use the ImageNet-1K-fine-tuned checkpoint reproduces its own nearest-class-mean ablation to within a point and closes that confound. One gate was still weaker than the rest and is now fixed. A consolidation used to face real held-out data only on the last task, the other classes being represented by contract samples alone; it now also faces the escrow the arbiter gate uses, over the whole validation split and every class seen so far. Replaying all 67 cached organisms through the old rule and the new one shows the old gate was not being misled: none of the 57 decisions that reach the escrow has negative overall evidence (median McNemar $z = +13.4$), so every consolidation already on record is corroborated by real data. The escrow adds exactly one refusal, where a class falls from $24/26$ to $17/26$ validation samples: refusing costs 12.2 points of test accuracy and protects that class, which is the trade the doctrine asks for. Transplanting the escrow verbatim first produced three false refusals, because its per-class hard floor was a fraction with no condition on how many samples the breach represented; on a benchmark with 120 classes at a median of 14 validation samples a three-sample swing tripped it. The floor now requires at least five lost samples, which leaves the collapse it exists for catchable and had never fired in the 41 arbiter decisions on record. Against the unconstrained run, certification costs 1–8 points on eight of twelve benchmarks (spoken digits -0.082, CIFAR-ViT -0.041, digits -0.036, MNIST -0.022, shapes -0.015, CIFAR-R18 -0.013, satimage -0.009, KMNIST -0.007), leaves three unchanged, and *improves* wall-robot by $+0.114$, because the unconstrained run had accepted a catastrophic arbiter there. Fifteen of 36 arbiters are accepted, zero accepted changes regress, and the split-MNIST scoreboard against same-data replay reads five of twelve (down from six under the unconstrained arbiter; certified digits 0.912 no longer clears replay's 0.933). Certification costs real accuracy on most benchmarks and buys it back only where the unconstrained version was badly wrong—which is the argument for it. Class order is not innocent: two random permutations of the task order move the certified numbers by 0.4–4.8 points on the MLP benchmarks (most on wall-robot and Fashion, least on 13-task letter), with the default order best on some and worst on others, so single figures on the small-task benchmarks carry a ± 2–4 point order band. Figure 1 shows every benchmark; Appendix A tabulates them.

7 Three findings, one mechanism

Three arcs of this study—redundant growth, market-based routing, and market-based contract renegotiation—end in laws that turn out to be the same statement seen from different sides. The acceptance gate's verdicts reproduce all three without being told them (Figure 2).

Routing headroom bounds every router. The gap $\mathrm{arbiter} - \mathrm{cascade}$ measures what any re-routing mechanism can win. It is benchmark-specific: $+0.001$ on KMNIST (first-accept already *is* the optimal router there, so re-routing by price can only destroy the birth-order prior), $+0.082$ on digits, $+0.062$ on CIFAR-ViT. Where headroom exists, a market with pool-calibrated prices takes most of it and can tie the central arbiter (0.836 ± 0.002 vs. 0.835 ± 0.003 on consolidated CIFAR-ViT, with no central trained state); where it does not, no pricing scheme helps. The gate enforces this unknowingly: on both evaluation and confirmation seeds it accepts arbiters $3/3$ exactly where *contract-covered* headroom exists (digits, CIFAR-R18, CIFAR-ViT, spoken digits), refuses where headroom is ≤ 0,

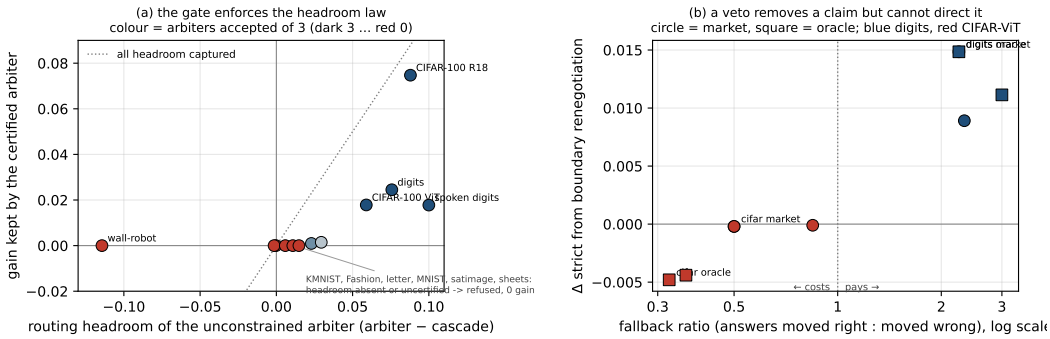


Figure 2: Two of the three laws. (a) The gate's verdicts reproduce the routing-headroom law without being told it: arbiters are accepted exactly where contract-covered headroom exists (upper right), refused where headroom is absent or negative, and refused where it exists but lives outside territory (the cluster at zero). (b) Boundary renegotiation pays only where the cascade's fallback is good: the fallback ratio separates the digits gains from the CIFAR-ViT losses, and even an oracle given the labels loses on CIFAR—a veto removes a claim but cannot direct it.

and refuses where headroom exists but lives outside territory (satimage, the sheets: the covered share is $\leq 5\%$).

Duplicate organs buy coverage, never precision. Growing the same class split twice and concatenating looks like a free ensemble; it cannot be one, because the first-accept prefix wins every contest. Measured over $k = 1 \ldots 4$ independent organisms (digits, five seeds): precision on covered inputs drifts $0.929 \to 0.922$ while coverage climbs $0.934 \to 0.981$, with zero flips against the prefix at every k—the entire $+0.038$ is abstention-filling, with sharply diminishing returns (the marginal twin answers its new inputs at 0.854, then 0.717, then 0.417). Width is not a substitute (0.872 at 59% more parameters than the twins' 0.905). The one contract-covered ensembling gain on the twin pool—majority vote among accepting organs, $+0.010$—is a *null* on every grown organism (0 answers changed in 30 runs across ten benchmarks, at contest rates up to 0.84 and up to 13 organs): organs that co-accept an input agree on its label, because vetoes leave only the generalist and the true specialist accepting. Voting had something to route only where competence was duplicated, and a grown anatomy never duplicates it. Redundancy and the unconstrained arbiter are *substitutes* (stacking adds $+0.004 \pm 0.006$: both climb to the same $\approx$0.96 ceiling by different routes), but redundancy and *certified* arbitration are *complements* on digits (0.946 fully covered, 72% of the unconstrained gain), because redundant coverage removes exactly the channel —filling—that certification forbids. On the flagship the complement vanishes (32% kept) and the remedy is topological: contracted prototypes. A by-product settles organ death: on this *grown* anatomy a sound per-class death certificate retires nobody (the anatomy is minimal), while on a twin anatomy it retires exactly one of each pair (5.0 ± 0.0 of 10 on KMNIST at a cost of 0.003)—so "no deaths" is a fact about the anatomy, not the gate (§8).

Selection cannot be replicated by monotone boundary edits. Since the unconstrained arbiter's power is choosing *who* answers, we asked whether its gain could be moved inside the certificate by renegotiating the territories themselves: organs bid pool-calibrated prices over the shared contract pool, and whatever an organ loses is removed by a veto, so the renegotiated contract is still a finite checkable spec. On digits this works and the **market matches an oracle given the labels** ($+0.0148$ both, fully certified, no inference-time trained component: what ships is a redrawn spec). On CIFAR-ViT even the oracle *loses* 0.0045: a veto removes a claim but cannot direct it—renegotiation is subtractive, so first-accept hands each ceded point to the next accepter in birth order, not to the bid winner. The *fallback ratio* (answers moved right against moved wrong) tracks the outcome monotonically across every configuration (3.00/2.25 on digits, gains; 0.76/0.35 on CIFAR, losses) and is computable on the contract pool *before* committing to any change. Two corollaries from the pricing arc: prices need joint *data* but not a central model (self-priced routing is anti-calibrated, Brier 0.31–0.71; calibrating the same per-organ curves on the shared pool improves Brier up to $200\times$); and

the auction needs a generalist *participant*, not a price index—adding the consolidated organ as an ordinary bidder is worth $+0.09$ to $+0.16$ everywhere (a gate-refused generalist scoring 0.187 alone still delivers $+0.092$ as a bidder: answering and pricing are different jobs), while indexing the reserve to its bid adds nothing.

8 The daring program

Five speculative mechanisms under the same frozen protocol on evaluation seeds: two work, one resolves into a positive claim about the anatomy, and two fail into the laws of §7.

Sleep cycles work if the organism dreams. Nightly gated consolidation plus a nightly arbiter rehearsal matches the one-shot lifecycle (digits 0.950 vs. 0.948; CIFAR-ViT 0.813 vs. 0.835) while adding the *anytime* property and structural immunity to mid-life failure. Dreamless sleep costs 4–6 points: an eagerly consolidated organ front-runs the cascade and starves new organs; the rehearsal compensates.

Apoptosis: the leak was per-class, and a sound certificate retires nobody. Gated organ death with the contract archived to a *genome* (which provably still powers later consolidations) retires 7–9 of 10 organs at 4–$8\times$ compression—but accuracy leaks ≈ 8 points through four escalating certificates (local synthetic, globally anchored synthetic, real escrows of 200 and $1{,}000$ samples; the last two identical, so the global check never binds). A fifth certificate adds *per-class* floors on a real escrow: **zero deaths, zero leak**. The diagnostic pins the mechanism: for a typical candidate every mean-based gate approves (coverage kept 0.97–0.99, global accuracy $+0.001$) while a single class drops 6.7–13.3 points; 19 of 20 refusals cite the per-class gate. The earlier leak was real damage invisible to means; the anatomy is minimal (no organ redundant at per-class resolution); and the twin control of §7 shows the certificate fires when redundancy is real. Death is the most expensive certificate in the lifecycle, and its price is per-class held-out data.

The market arc (compressed; laws in §7). Self-priced decentralized routing fails by anti-calibration; a market-maker calibrating the same per-organ curves on the shared contract pool repairs the prices and, after consolidation, ties the central arbiter on the flagship; used at contract-formation time it matches an oracle on digits and cannot help on CIFAR, where selection, not boundaries, is what the cascade lacks.

Second-order contracts: an SLA on the organism's own drift. Before each change event the system publishes a bound (its contract-sampled forecast times an online conformally calibrated factor, $\alpha{=}0.25$, preregistered). Over 54 events: 5 violations (9.3% against the 25% budget) at median tightness 1.1–$1.45\times$; online calibration beats the fixed prior 5 vs. 14.

The fractal organism composes. A super-organ wraps a whole sub-organism behind one merged territory (a union of PCA territories is a PCA territory) and presents the standard organ interface, top-level vetoes included. Ten tasks in three super-epochs: strict 0.718 ± 0.003, top-level growth flips exactly 0 (the invariant survives nesting), and an entire $\sim$30-class super-organism re-derived from its merged contract alone agrees at 0.75–0.78 on its jurisdiction.

9 The feature ledger

Every remaining mechanism isolated through the full pipeline on both flagships (evaluation seeds; removal rule preregistered). Nothing is removed: vetoes are the backbone (removal: -0.19 MNIST, -0.50 CIFAR-ViT—the arbiter does *not* subsume inhibition; without vetoes growth itself drowns at 0.10–0.13), the spectral cap earns its keep (-0.02/-0.07; uncapped evidence also breaks the arbiter gate on all six runs), and adolescence adds $+0.09$ beyond the arbiter (consolidation repairs representation; arbitration repairs routing). Previously measured and excised: scalar tightening (-0.08 when added back), trunk-mode consolidation, mid-life timing (-0.05), renegotiated territories (vacuous), self-priced routing, the unconstrained arbiter (uncertifiable).

10 What fails, and why

Weak features break the cascade. With 100 fine-grained classes on frozen ResNet-18 features, territories overlap; vetoes halve rather than erase false accepts; NCM wins (0.55 vs. 0.39). The regime-appropriate topology is prototypes; the contract machinery ports to it at zero cost, the cascade

does not. **Gates are certificates relative to contract fidelity.** On wall-robot the unconstrained arbiter passed both original gate signals yet regressed ≈ 12 points on test, systematically, on all three evaluation seeds: the sampler misrepresents old tasks and contract samples cannot see it. The repaired gate refuses it—by escrow on real data, decisively, on evaluation and confirmation seeds alike—but the episode's lesson stands: detecting a real-data regression requires real held-out data, and the gate's power is bounded by the escrow it is given (`detectable_drop` states the bound). **The spoken-digits benchmark leaks speaker identity.** All FSDD numbers above use the standard random split, which shares the same six speakers across train, validation and test. Holding one speaker out for test and another for validation, *every* method collapses—organism $0.772 \to 0.349$ (certified lifecycle 0.328), replay $0.93 \to 0.45$, the joint model $0.95 \to 0.52$—so the collapse belongs to the features and data (an 8×16 log-spectrogram, four training speakers), not to the organism, and relative standing is preserved (-0.10 to replay against -0.16 before). The same experiment exposes the precise scope of the acceptance gate: on one seed it *accepted* a -0.062 test regression at escrow $z = +4.91$, because the arbiter genuinely improved on the validation speaker and test is a different speaker. **The gate certifies non-regression on the distribution its escrow samples**; it cannot certify generalisation to one it has never seen, and where deployment differs from validation along a known axis the escrow must be held out along that axis. **Renegotiated territories are vacuous** when refitted from generated data: they pass every audit and answer nothing. **Sampler texture.** On raw pixels every sampler is trivially discriminable from real data; consolidation still transfers, but adolescence gates correctly refuse where it does not (MNIST/Fashion pixels; 8 of 8 refusals justified, 6 of 6 acceptances beneficial across the study). **The unconstrained arbiter is uncertifiable**, not mis-gated: 45–94% of its advantage is bought outside every certified territory, and no monotone repair (gating, boundary renegotiation) recovers more than $72\%/32\%$ of it (§4, §7). **Scale.** MLPs, a $\leq$160-unit sheet, $\leq$100 classes, 2–3 seeds; baselines are same-size reimplementations; CIFAR gate-confirmation seeds outstanding.

11 Discussion

Continual learning is scored on accuracy; deployed systems are also governed by change control. The certified organism's property set—verified zero forgetting, abstention semantics, finite replaceable specifications, gated consolidations with an out-of-sample-confirmed acceptance gate, every answer contract-covered, logged and self-forecasting drift—is a change-control discipline for learning systems, obtained at replay-level accuracy without stored data. Certification is not free: it costs 1–8 points against the unconstrained lifecycle on most benchmarks and buys them back only where the unconstrained system was badly wrong—which is precisely the case change-control exists for. The recurring methodological lesson of the study is the parents' lesson compounded: measure the suspected bottleneck before fixing it (the sampler was innocent; a training pathology and a miscalibrated gate were guilty), make every certificate carry a control that must fail (the twin anatomy, the swapped spec, the discrimination audit), and price every guarantee.

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A The certified lifecycle on all benchmarks

Regenerated from `results/lifecycle_gated_full.json`. Stages: growth → adolescence (gate may refuse; refused stages carry forward) → territory-respecting arbiter under the repaired gate. Strict accuracy, evaluation seeds 2–4; the unconstrained arbiter's score on the same organism is shown as the uncertified reference.

benchmark	growth	+ adolescence	**+ certified arbiter**	unconstr. (uncert.)
split-digits	0.877	0.888 (1/3)	**0.912** (3/3)	0.964
split-MNIST	0.892	*refused*	**0.893** (2/3)	0.915
split-KMNIST	0.805	*refused*	*refused*	0.805
split-Fashion	0.799	*refused*	*refused*	0.797
letter (13 tasks)	0.782	*refused*	*refused*	0.788
satimage	0.832	*refused*	**0.833** (1/3)	0.861
wall-robot	0.800	*refused*	*refused*	0.685
CIFAR-100 (frozen R18)	0.395	*refused*	**0.470** (3/3)	0.483
CIFAR-100 (ViT-B/16)	0.667	0.776 (3/3)	**0.794** (3/3)	0.835
sheet: sweeps	0.979	0.989 (3/3)	*refused*	1.000
sheet: shapes	0.985	*refused*	*refused*	1.000
sheet: spoken digits[†]	0.772	*refused*	**0.790** (3/3)	0.872

[†]Shared-speaker split; speaker-disjoint: growth 0.349, certified 0.328 (§10). Refusals leave the organism unchanged, so every "refused" row keeps its previous stage's number and guarantee. Wall-robot's harmful arbiter (candidate 0.646–0.710) is refused on all evaluation and all confirmation seeds; the unconstrained run had accepted it at 0.685. Confirmation on untouched seeds 5–7 (ten CPU benchmarks, 30 runs): 9 accepted, zero clearly-helpful refused, one accepted regression of -0.0068—half the gate's stated resolution for that escrow.

B Claim-by-claim evidence map

Every quantitative claim above is regenerated by `claims.py` into `results/claims.json`. Keys: retention counts (`B2_*.growth_flips_total`, `B_sheet.*`); veto measurements (`veto_measurement`); the tightening ablation (`ablate_tightening_mnist`); OOD geometry (`OOD_geometry`); class-IL tables (`B2_continual_*`); CIFAR-100 (`B_cifar100*`); drift audits (`B_drift_*`); adolescence and renegotiation (`B_adolescence`, `B_renegotiate`, `B_cifar_adol_*`); the certified lifecycle and its confirmation (`B_lifecycle_gated`, `B_lifecycle_confirm`); the gate and its audit (`B_gatefix`, `B_gate_audit`); the arbiter decomposition (`B_arbiter_source`, `B_arbiter_cifar`); redundant growth (`B_redundant`, `B_ensemble`); the market arc (`B_market`, `B_market_maker`, `B_market_indexed`, `B_negotiate`); apoptosis (`B_apoptosis`, `B_apoptosis_v4`); sleep (`B_sleep_dream`); the drift SLA (`B_meta_sla`); the fractal organism (`B_fractal`); the feature ledger (`B_isolate`); sampler fidelity (`B_sampler`); replacement (`B_replace_digits`); the sheet audit (`B_sheet_audit`).